

Return/Dispute Case: AMD EPYC 9B14

Suspected Defective CPU — Evidence Package

Host: zappa2 | CPU: AMD EPYC 9B14, 96c/192t (OEM-exclusive SKU) | Generated 2026-10-01

Status: CPU return not yet filed with the original seller — a defect claim was filed, but a bank/card dispute is being prepared as a fallback if the seller does not resolve it. This document is the evidence package for that dispute.

Scope note (2026-10-01): This case rests on two distinct phenomena, and only one is proven localized to the CPU. The corrected SERR/PERR pattern is confirmed and localizes cleanly. The silent hangs — the failure that actually costs rentals/uptime — currently have **zero** AER/MCE/extlog signature across three occurrences, despite OS-managed AER (pcie_ports=native) active and rasdaemon running throughout. The hangs cannot yet be attributed to the CPU, or to PCIe at all, by log evidence. The CPU swap test is the right next step — it is mechanism-agnostic — but this filing does not claim the hangs as CPU-localized evidence until that test confirms it.

What Is Confirmed vs. Not

Phenomenon	Evidence	Status
Corrected SERR/PERR burst at cold boot	Localizes to CPU root complex (00:01.4 / 00:01.5), reproducible across 3 boards	Confirmed, CPU-consistent. Independently corroborated as benign link-training noise, not itself causing outages.
Silent host hangs (the actual outage / reliability2 impact)	rasdaemon DB (ras-mc_event.db) checked 2026-10-01: no new entries since 2026-09-03 — neither the 08:03 nor ~13:50 hang produced an AER/MCE/extlog record, despite rasdaemon running and pcie_ports=native active	Not confirmed as a PCIe or CPU-root-complex event. A watching kernel that logs nothing during the hang is evidence no PCIe error occurred at the point of failure — found once before on board #2, reconfirmed independently on board #3.

Practical consequence: the "why the CPU" argument is solid for the SERR/PERR pattern itself. It is **not** solid for the hangs — those remain in the harder-to-diagnose bucket (power delivery, socket/CPU seating, IO-die silicon) that leaves no log trail at all. The swap test is the only evidence that will actually settle the hangs, regardless of mechanism.

The Claim, Stated Plainly

The same corrected PCIe fault signature (SERR/PERR bursts on the CPU's own IO-die root complex) has now been observed across **three different motherboards**. The motherboard has been replaced twice and the fault persists. The CPU is the only major component that has remained constant across all three builds. By elimination for this specific signature, the motherboard is cleared and the CPU is the leading suspect. The separate question of what is causing the silent hangs is still open and is being settled by the CPU swap test, not by this log-based argument.

Why the CPU, Not the Boards

- Three motherboards, one fault signature.** If any single board's NVMe slot, chipset, or traces were the root cause, replacing the board should have cleared it. It did not, twice in a row (board #1 → #2 → #3, same class of PCIe error each time — GPU-slot SERRs on board #1, NVMe-port SERRs on board #2, recurrence again on board #3 as of 2026-10-01).
- The fault localizes to the CPU's own root complex, not to a device.** On board #2, every SERR landed on 00:01.4 or 00:01.5 — sibling functions of the same Genoa IO-die root complex — not independent slots. Swapping which NVMe slot was populated moved the fault between siblings; it never went away. That pattern points upstream of any one device, to the CPU's PCIe controller.

- 3 **2026-10-01 evidence reinforces this, it does not point at the drive.** `lspci -vv -s 03:00.0` on the current (3rd) board shows the primary NVMe — a Crucial T705, rated and negotiating LnkCap: 32GT/s x4 — currently training down to LnkSta: 32GT/s x2 (downgraded). On this class of platform the primary M.2 slot is wired directly to the CPU's own PCIe root complex, not through the chipset. A link-width downgrade like this, recurring across independently-wired slots on independent boards, is the signature of a marginal source-side (CPU) lane or SerDes, not a drive defect — the drive's SMART health has been confirmed pristine throughout.
- 4 **The hangs themselves are not yet attributable to the CPU** — flagged here for honesty, not to support the claim. On board #2 and again on board #3 (2026-10-01), a host hang occurred with the BMC fully responsive, all rails nominal, no SEL entry, kernel alive (serving ICMP, printing D-state task dumps) but unable to complete disk I/O — and no AER/MCE/extlog record at all, confirmed via rasdaemon's own persisted database. That absence of a PCIe signature means the hang cannot currently be pinned on the CPU's IO die specifically, or on PCIe as the failure path at all. It remains possible, but unproven — the swap test is what will actually decide it.
- 5 **The PERR burst pattern is consistent across all three boards** — ~14 Critical Interrupt #0x80 | PCI PERR events in the BMC SEL, all within the same second, at every cold boot. Identical signature, three independent boards, three independent BMCs.

What Has Been Ruled Out

- **NVMe drive defect** — SMART pristine, current firmware, no media errors in `nvme error-log`. The one hang captured in detail decoded to NVME_SC_HOST_ABORTED_CMD (transport/path error), not a media or data integrity status — the drive never reported bad data; the host gave up waiting on it.
- **Motherboard (both prior boards)** — same fault class survived both replacements.
- **PSU/ground offset** — considered and partially tested on board #2; does not explain why the fault localizes to one root complex rather than scattering across every GPU link that would span a ground-domain split.
- **Thermal cause** — the hottest documented period of operation (intake air at 50°C, 8 GPUs under full load) produced zero faults; all faults so far have occurred cool and idle or under light load.
- **Software/workload cause** — one hang was independently identified as a memory-exhaustion livelock and fixed (swap disabled, earlyoom armed); explicitly excluded from the CPU claim and not part of this case.

Timeline (Condensed)

Date	Board	Event
(pre-09/12)	#1	SERR attributed to GPU slots 0/1
09/12–09/20	#2 (TYAN S8056GME)	Multiple SERR bursts on 00:01.4/00:01.5; 3 silent hangs, 1 explained as memory livelock, 2 unexplained
09/21	#2	AER baseline captured (14 endpoint / 1 root-port corrected errors, idle)
~09/25	#3	Board swap #2, CPU reseated with corrected torque
10/01 08:03	#3	Hard reboot. BMC SEL: 14x PCI PERR burst immediately preceding reboot. Kernel AER log: device_id 0000:03:00.0, corrected, same signature as board #2
10/01 11:22	#3	<code>lspci -vv -s 03:00.0</code> shows NVMe link downgraded to x2 (rated x4)
10/01 ~13:50	#3	Second silent hang, same day as the 08:03 crash. Pre-crash kernel log ends cleanly at 13:45:43 on routine Docker veth churn, then nothing. No panic, no AER, no OOM, no soft lockup. ~8 min gap before power-cycle via BMC from zappa1. Boot-time PERR burst on recovery boot (13:53:51-52) is known-benign link training, not a separate fault.

Frequency note: two independent hangs in one calendar day (08:03 and ~13:50) is a step up from the prior cadence of roughly one every few days. If this pace continues, it strengthens the case that something is actively degrading rather than static — but per the scope note above, the hangs themselves have no log signature yet, so this is evidence of worsening frequency, not evidence of which component is degrading.

Evidence to Attach

- TROUBLESHOOTING-PCIE-SERR.md (full technical record, board #2)
- BMC SEL export covering the 10/01 08:03 PERR burst (ipmitool sel elist)
- Kernel log excerpt, 10/01 08:03:22, device_id: 0000:03:00.0 corrected AER records
- lspci -vv -s 03:00.0 output, 10/01 11:22, showing the x4 → x2 downgrade
- diagnostics output bundle for the 10/01 08:03 incident (uptime, GPU state, fault watcher section)
- sel-elist-2026-10-01_1357.txt — clean BMC SEL export covering the second 10/01 hang (~13:50) and its recovery boot
- journalctl -k -b -1 excerpt for the ~13:50 hang, showing the silent cutoff at 13:45:43
- ras-mc-ctl --errors output and /var/lib/rasdaemon/ras-mc_event.db state (2026-10-01) showing no AER/MCE/extlog entry for either hang — documents the scope limit above, attach alongside the SERR evidence, not as evidence for the CPU claim
- Photographs of CPU socket seating/torque at each of the 3 builds, if available

Replacement CPU Ordered — The Decisive Swap Test

Ordered 2026-10-01. AMD EPYC 9B14, unlocked (100-000000782), from eGoods Supply (ebay.com, 25,784 feedback, 99.9% positive), \$1,599, used condition, expected delivery Oct 3-6. This is being bought to run the decisive test: swap only the CPU, keep board #3, the same Crucial T705 NVMe, and the same seating/torque procedure. Do not dispose of or return the current CPU until this test is complete and the dispute is resolved — it is the physical evidence for the claim.

Procedure, once the new CPU arrives:

- 1 Capture a final AER baseline and full diagnostics output on the current CPU immediately before teardown, for a clean before/after.
- 2 Change only the CPU. Same board, same NVMe, same slot, same SP5 torque sequence used on the board #2→#3 rebuild.
- 3 First boot: capture AER baseline (aer_dev_correctable on 03:00.0, aer_rootport_total_err_cor on the root port) at comparable idle uptime, same method as the 2026-09-21 baseline.
- 4 Run the rig normally and watch for recurrence over the following days.

Reading the Result

Outcome	Verdict
Fault clears on the new CPU, same board	CPU confirmed defective. Also exonerates board #3 (and retroactively boards #1/#2). Strongest possible evidence for the dispute — attach this test's before/after logs directly.
Fault persists on the new CPU, same board	CPU is cleared. Points back at board #3 or a rig-level cause (PSU ground offset). Do not proceed with the CPU dispute on weak grounds if this happens — the Mitac board case becomes the live one instead.

Open Items

- Confirm with seller/dispute whether the same physical CPU was moved across all 3 motherboards (this is the load-bearing assumption of the whole case — verify and state explicitly in the dispute filing)
- Attach this file + the evidence list above to the bank dispute if the seller does not resolve the existing defect claim — state the SERR vs. hang distinction explicitly rather than presenting both as equally CPU-attributed
- DONE: Decide whether to pursue CPU replacement in parallel — ordered, see above
- Run the swap-test procedure once the new CPU arrives (Oct 3-6) and record the outcome — this is now the only evidence that will settle the hangs, since log-based localization doesn't reach them
- If the swap test clears the CPU (fault persists), do not retroactively claim the hangs as CPU evidence anywhere this file has been referenced or attached